

ST207 Group Project - AT2024

Group: Weather_db

January 31 2025

Candidate Numbers:

39838

47390

48372

49708

Our focus is on historical, though recent, weather trends in Southeastern England to identify the best possible farming practices. The weather affects a variety of decisions farmers need to make regularly. In the long term, climate affects where farmers choose to live, what large equipment they should purchase, and, on a macro-level, what areas land-use planners appropriate to farming. However, we decided to focus on the shorter-term decisions made frequently by farmers. These include but are not limited to: what crops are most viable and/or profitable for a farm in a given area in a given season/year, what seed type is best considering a specific area's weather, or how many harvest cycles will the weather allow for in specific time-frame.

One main constraint of this topic is that other factors affect these decisions as well, most of all soil types. Weather is immensely important for farming decisions, but it is not necessarily reliable enough on its own for the kinds of use cases we hoped to explore. While it could be argued that we should have collected soil type data alongside our weather data to allow it to contain more insightful relationships, this would have required significantly more data storage (as soil type data requires much more geographic specificity to be useful than weather data) and, frankly, we do not have enough understanding of agricultural biology to confidently draw accurate conclusions from combining weather and soil type data anyways.

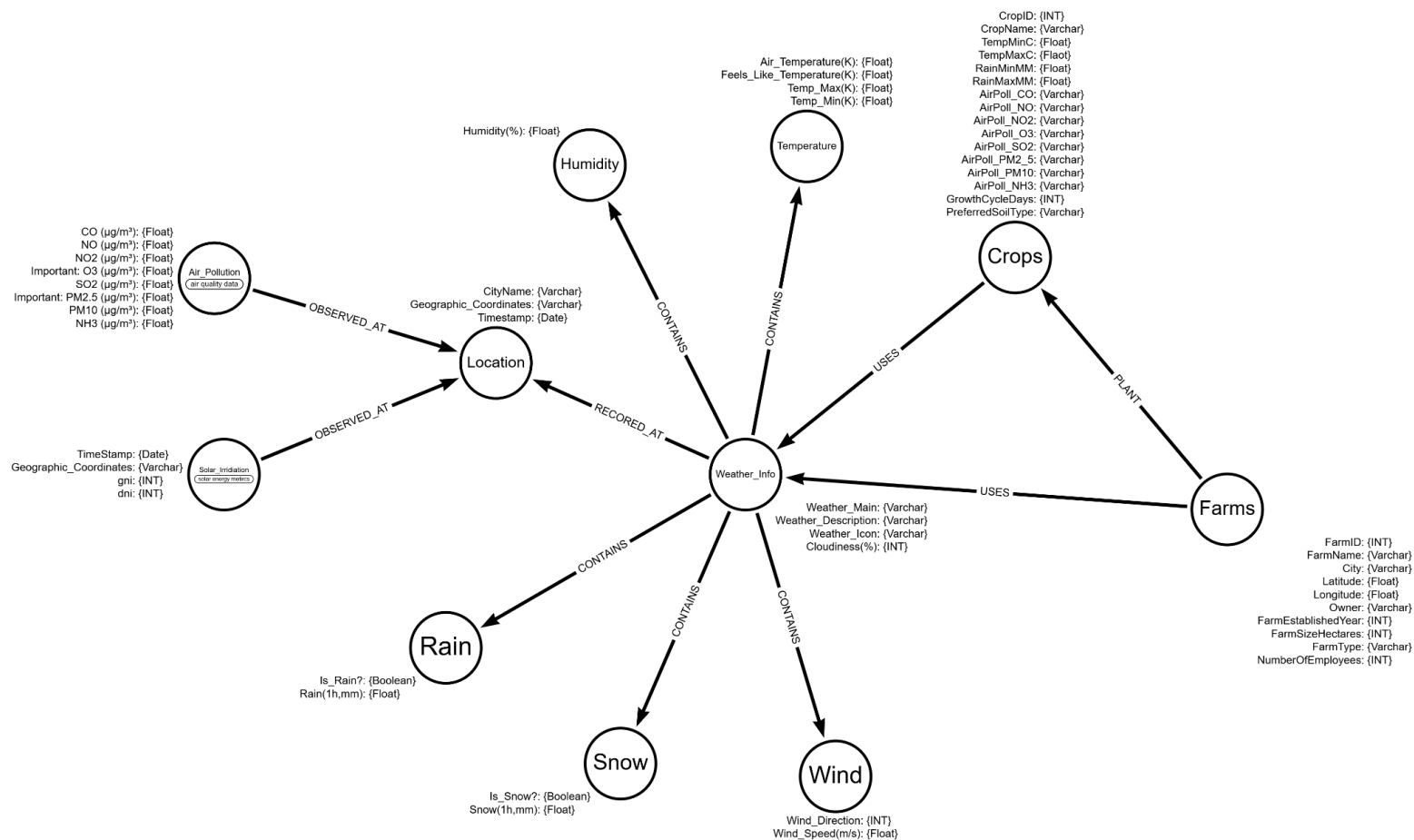
In the real world, farms, location information, and crops' values to farmers are all entities, while weather information is an attribute (with many sub-attributes) of the location information. In terms of relationships, farms use weather information and crops' values both use weather information. There are no generalisations or specialisations dependencies.

One advantage of weather data is an abundance of available standard, tabular, weather data available, so there is no need for collecting non-standard data for this topic. We were able to obtain over fifty thousand measurements per location for just solar irradiance information (from Solcast) alone. After gaining access to OpenWeather API's Developer Pack, we found that it was difficult to comb through the vast amount of available centralised data to find which metrics were valuable instead of having difficulties finding reliable weather data. Solcast and OpenWeather both appear to be reliable sources of information and, collectively, they covered more than enough for us to draw from in terms of weather data.

We found it more difficult, however, to find applicable and accessible data regarding the implications of weather on crops (and other things to apply weather data to in general). After looking at several data sources about crops, we found that there are only limited data sources available that match our requirements. This includes many famous data sources:

- DEFRA API
 - Data provided by UK GOV open data plan
- FAO
 - Food data information provided by the United Nations
- Numerous other APIs including CROME, OpenWeather Agriculture API, etc.

The most common issue is they do not provide options for bulk data collection, with only some exceptions for specific groups. For instance, the DEFRA is a good data source, but it only provides full access to certain government agents and scholars. Even farmers are limited to accessing data relevant to their property only. Therefore, we decided to use ChatGPT to generate synthetic data. Both the farm information and the crop information are generated with AI. We believe synthetic farm information is ok, as our focus is on the effect of weather on farmers' decisions, not to see what type of farm, for example, is best for specific weather conditions, because we see that as more appropriate in a long-term climate study, rather than a sample of weather data in a relatively narrow range of dates. It would have, however, been preferable to access better data regarding the relationship between crop viability and weather conditions to be able to draw more meaningful conclusions.



This is the graphical conceptualisation of our project which is interpreted by using arrow.app. The graph shows all entities and attributes of our database, and all relationships between them. We have also specified the data type for every attribute, aiming to give users a better understanding of our database. Our target users are the working staff of farms, and the project aims to give recommendations about crops to them. This is the major reason why we choose to conceptualise a database using a graphical approach instead of an ER Diagram: our target users are not always experts in database manipulation. Our diagram spots where the farms and crops are compared to an ER Diagram, and what factors of farms and crops (e.g. sunshine time, amount of rain, temperature) we have considered for them. Therefore, farmers will have a rough understanding of what factors influence recommendations derived from our database. Transparency in the recommendation process would broaden its support.

Core Entities and Their Attributes

1. Location

o Attributes:

- *CityName* (Varchar)
- *Geographic_Coordinates* (Varchar)
- *Timestamp* (Date)

o **Purpose:** Represents a specific place at a particular point in time where data (e.g., weather conditions, air quality) is observed or recorded.

2. Air_Pollution

o Attributes:

- *CO, NO, NO2, O3, SO2, PM2.5, PM10, NH3* (all Float, in $\mu\text{g}/\text{m}^3$)

o **Purpose:** Captures air quality metrics critical for agricultural health and environmental assessments.

3. Solar_Irradiation

o Attributes:

- *TimeStamp* (Date)
- *Geographic_Coordinates* (Varchar)
- *gni* (INT), *dni* (INT)

o **Purpose:** Holds solar energy-related measurements (Global Normal Irradiance and Direct Normal Irradiance). Useful for determining how much sunlight crops might receive, planning solar panel usage, etc.

4. Weather_Info

o Attributes:

- *Weather_Main* (Varchar)
- *Weather_Description* (Varchar)
- *Weather_Icon* (Varchar)
- *Cloudiness (%)* (INT)

o **Purpose:** Encapsulates general weather information (e.g., sunny, cloudy, stormy). It is further detailed by associations with *Temperature*, *Humidity*, *Rain*, *Snow*, and *Wind*.

5. Temperature

o Attributes:

- *Air_Temperature(K)* (Float)
- *Feels_Like_Temperature(K)* (Float)
- *Temp_Max(K)* (Float)
- *Temp_Min(K)* (Float)

o **Purpose:** Tracks thermal conditions relevant to weather forecasts and crop requirements.

6. Humidity

- o **Attribute:**

- *Humidity(%)* (Float)

- o **Purpose:** Records moisture levels in the air, essential for understanding plant and soil water needs.

7. Rain

- o **Attributes:**

- *Is_Rain?* (Boolean)
 - *Rain(1h, mm)* (Float)

- o **Purpose:** Indicates rainfall occurrence and volume (useful for irrigation planning and flood risk analysis).

8. Snow

- o **Attributes:**

- *Is_Snow?* (Boolean)
 - *Snow(1h, mm)* (Float)

- o **Purpose:** Tracks snowfall occurrence and volume, relevant to cold-weather crop planning and farm logistics.

9. Wind

- o **Attributes:**

- *Wind_Direction* (INT)
 - *Wind_Speed(m/s)* (Float)

- o **Purpose:** Provides data on wind patterns, speeds, and directions, useful for farm safety, drone usage, or pesticide spraying decisions.

10. Crops

- o **Attributes:**

- *CropID* (INT)
 - *CropName* (Varchar)
 - *TempMinC, TempMaxC, RainMinMM, RainMaxMM* (Float)
 - *AirPoll_CO, AirPoll_NO2, AirPoll_O3*, etc. (Varchar)
 - *GrowthCycleDays* (INT)
 - *PreferredSoilType* (Varchar)

- o **Purpose:** Catalogs different crop types, along with their temperature, rainfall, and air quality tolerances, and overall growth requirements.

11. Farms

- o **Attributes:**

- *FarmID* (INT)
 - *FarmName* (Varchar)
 - *City* (Varchar)
 - *Latitude, Longitude* (Float)
 - *Owner* (Varchar)

- *FarmEstablishedYear* (INT)
- *FarmSizeHectares* (INT)
- *FarmType* (Varchar)
- *NumberOfEmployees* (INT)

o **Purpose:** Represents actual farms, storing both their geographic data and administrative details.

Key Relationships

- **Location ↔ Air_Pollution:**

o *OBSERVED_AT* indicates that air pollution metrics are tied to a specific place and time.

- **Location ↔ Solar_Irradiation:**

o *OBSERVED_AT* similarly links solar data (GNI, DNI) with a particular location and time.

- **Location ↔ Weather_Info:**

o *RECORDED_AT* shows how broader weather information (like “Partly Cloudy,” “Rainy,” etc.) is logged in association with a specific location and timestamp.

- **Weather_Info ↔ Temperature / Humidity / Rain / Snow / Wind:**

o *CONTAINS* highlights how a single weather event or “Weather_Info” record can include temperature details, humidity levels, whether it’s raining or snowing, and wind data.

o These relationships unify separate specialized measurements (temp, wind, etc.) under one overarching weather snapshot.

- **Farms ↔ Crops:**

o *PLANT* indicates which crops are grown on which farms, forming the agricultural foundation of the system.

- **Weather_Info ↔ Crops:**

o *USES* represents the idea that crops (and the farmers growing them) depend on or “use” weather information. For example, temperature and rainfall data help determine sowing times, fertilizer usage, or harvest schedules.

- **Farms ↔ Weather_Info:**

o *USES* similarly suggests that farms consult or rely on weather data to manage daily operations (irrigation, pesticide application, etc.).

Queries:

No updating queries were used as historical data is inherently static.

1. Calculating **total** rainfall, in mm, in all eight locations in 2024, for the whole year, each season, and each month

Total rainfall (mm) in location in time periods:																	
Location	Year	Winter	Spring	Summer	Autumn	January	February	March	April	May	June	July	August	September	October	November	December
Buckinghamshire	1491.68	448.78	358.44	192.08	492.38	122.90	262.38	127.52	154.32	76.60	37.56	110.46	44.06	288.00	89.96	114.42	63.50
Location	AQI	CO	NO	NO2	O3 (Crucial/Most Harmful)			SO2	PM2.5 (Crucial/Most Harmful)			PM10	NH3				
Buckinghamshire	1.68262	221.72382	0.25923	3.95641				61.09361	1.08978				3.81290	3.80049	1.95040		
East Sussex	1.87476	227.37688	0.51023	6.18409				69.20285	2.59569				4.86573	5.34998	0.79891		
Essex	1.70583	232.46629	1.42064	9.33633				58.46953	3.10123				4.59944	4.82291	1.04413		
Hertfordshire	1.67429	231.28632	1.36484	9.79511				56.90881	3.09821				4.37097	4.58870	0.92469		
Kent	1.73893	225.82381	0.75194	6.02101				62.04775	1.96739				4.63156	4.99522	0.98865		
London	1.68821	250.90783	5.76573	17.29157				50.92735	7.52799				5.37116	5.63128	0.82823		
Surrey	1.71238	231.41025	2.05732	9.96325				58.57696	3.22558				4.30370	4.41637	0.83049		
West Sussex	1.75000	223.58687	0.49480	6.39981				62.90793	2.21789				4.44628	4.75808	1.06806		
Average	1.72838	230.57276	1.57809	8.61845				60.01685	3.10297				4.55022	4.79538	1.05420		

2. Calculating the annual **average** air pollution data for all cities in 2024

Location	AQI	CO	NO	NO2	O3 (Crucial/Most Harmful)	SO2	PM2.5 (Crucial/Most Harmful)	PM10	NH3
Buckinghamshire	1.68262	221.72382	0.25923	3.95641	61.09361	1.08978	3.81290	3.80049	1.95040
East Sussex	1.87476	227.37688	0.51023	6.18409	69.20285	2.59569	4.86573	5.34998	0.79891
Essex	1.70583	232.46629	1.42064	9.33633	58.46953	3.10123	4.59944	4.82291	1.04413
Hertfordshire	1.67429	231.28632	1.36484	9.79511	56.90881	3.09821	4.37097	4.58870	0.92469
Kent	1.73893	225.82381	0.75194	6.02101	62.04775	1.96739	4.63156	4.99522	0.98865
London	1.68821	250.90783	5.76573	17.29157	50.92735	7.52799	5.37116	5.63128	0.82823
Surrey	1.71238	231.41025	2.05732	9.96325	58.57696	3.22558	4.30370	4.41637	0.83049
West Sussex	1.75000	223.58687	0.49480	6.39981	62.90793	2.21789	4.44628	4.75808	1.06806
Average	1.72838	230.57276	1.57809	8.61845	60.01685	3.10297	4.55022	4.79538	1.05420

3. Get the **average** solar irradiance data for all cities in 2024

Location	DNI	GHI
Buckinghamshire	111.526071	125.099762
East Sussex	128.789286	135.524881
Essex	110.396786	127.948095
Hertfordshire	109.494167	125.339048
Kent	111.990238	128.328452
London	105.733690	121.177024
Surrey	103.352738	123.144405
West Sussex	123.433095	132.635476
Average	113.089509	127.399643

4. Get the total snowing days for all cities in 2024
- a. Snowing day is defined as days where Snow_1h_mm is non-zero at any point

Location	SnowDays
Buckinghamshire	7
East Sussex	2
Essex	6
Hertfordshire	4
Kent	3
London	3
Surrey	3
West Sussex	4
Average	4

5. Calculate annual/seasonal/monthly average daily Max/Min Temperature (°C) for all 8 cities in 2024
- a. We converted temperatures from Kelvin units (how MongoDB stores it) to Celcius, the most commonly used unit worldwide
- b. Monthly temperature date is to the right

=== Annual Temperature Data (°C) ===

Location	AvgTempMaxC	AvgTempMinC
Buckinghamshire	12.384239	10.232150
East Sussex	13.334062	10.823354
Essex	13.212248	10.657167
Hertfordshire	13.060337	10.536615
Kent	13.591689	10.851283
London	13.504560	10.822075
Surrey	13.361983	10.651054
West Sussex	13.285058	10.569517
Average	13.216772	10.642902

=== Seasonal Temperature Data (°C) ===

Location	AvgTempMaxC	AvgTempMinC	Season
Buckinghamshire	11.952001	9.970250	Autumn
Buckinghamshire	11.713909	9.384380	Spring
Buckinghamshire	17.664102	15.370519	Summer
Buckinghamshire	7.583807	5.631297	Winter
East Sussex	13.224460	11.042659	Autumn
East Sussex	12.430254	9.662763	Spring
East Sussex	18.499139	15.780782	Summer
East Sussex	8.569338	6.221676	Winter
Essex	12.822821	10.466193	Autumn
Essex	12.402939	9.694787	Spring
Essex	18.803245	15.941685	Summer
Essex	8.167491	5.917544	Winter
Hertfordshire	12.656480	10.300769	Autumn
Hertfordshire	12.263845	9.606925	Spring
Hertfordshire	18.636153	15.771060	Summer
Hertfordshire	8.034613	5.867928	Winter
Kent	13.312131	10.984752	Autumn
Kent	12.757912	9.676875	Spring
Kent	19.023931	15.974606	Summer
Kent	8.632216	6.173524	Winter
London	13.049449	10.653789	Autumn
London	12.729965	9.853166	Spring
London	19.011454	16.055727	Summer
London	8.591596	6.122659	Winter
Surrey	12.919138	10.500333	Autumn
Surrey	12.611780	9.691046	Spring
Surrey	18.789324	15.762856	Summer
Surrey	8.497859	6.061415	Winter
West Sussex	13.157865	10.685183	Autumn
West Sussex	12.415222	9.440091	Spring
West Sussex	18.417926	15.577931	Summer
West Sussex	8.537710	5.991767	Winter
Average	13.058752	10.494279	nan

=== Monthly Temperature Data (°C) ===

Location	AvgTempMaxC	AvgTempMinC	Month
Buckinghamshire	5.909681	3.813472	January
Buckinghamshire	8.996437	6.905977	February
Buckinghamshire	9.197218	7.036632	March
Buckinghamshire	10.727847	8.510431	April
Buckinghamshire	15.184852	12.518629	May
Buckinghamshire	16.176569	13.593069	June
Buckinghamshire	18.002141	15.862328	July
Buckinghamshire	18.787419	16.630551	August
Buckinghamshire	15.250187	13.088807	September
Buckinghamshire	12.358239	10.465094	October
Buckinghamshire	8.338957	6.439402	November
Buckinghamshire	8.069978	6.553961	December
East Sussex	6.909306	4.343153	January
East Sussex	10.030934	7.567299	February
East Sussex	10.110806	7.426976	March
East Sussex	11.614042	8.966611	April
East Sussex	15.539583	12.572245	May
East Sussex	17.048639	13.888167	June
East Sussex	18.764167	16.258707	July
East Sussex	19.654919	17.165255	August
East Sussex	16.179555	13.987629	September
East Sussex	13.871868	11.682177	October
East Sussex	9.693978	7.530139	November
East Sussex	8.958731	7.131926	December
Essex	6.400931	4.015208	January
Essex	9.677888	7.365718	February
Essex	9.931774	7.482298	March
Essex	11.466389	8.869847	April
Essex	15.790121	12.705605	May
Essex	17.323375	13.976722	June
Essex	18.935259	16.336882	July
Essex	20.118882	17.473562	August
Essex	16.274181	13.778549	September
Essex	13.58347	10.972823	October
Essex	9.031196	6.735549	November
Essex	8.650394	6.709125	December
Hertfordshire	6.262792	3.991875	January
Hertfordshire	9.534799	7.298420	February
Hertfordshire	9.781962	7.344785	March
Hertfordshire	11.349597	8.776931	April
Hertfordshire	15.630484	12.672285	May
Hertfordshire	17.220542	13.910583	June
Hertfordshire	18.789828	16.180374	July
Hertfordshire	19.862339	17.188616	August
Hertfordshire	16.071940	13.558003	September
Hertfordshire	13.085228	10.814556	October
Hertfordshire	8.966620	6.616078	November
Hertfordshire	8.541357	6.645033	December
Kent	6.940236	4.264542	January
Kent	10.120460	7.573233	February
Kent	10.347836	7.569220	March
Kent	11.924292	8.989014	April
Kent	15.974718	12.450202	May
Kent	17.570306	13.895014	June
Kent	19.197011	16.481422	July
Kent	20.268750	17.512997	August
Kent	16.566422	14.070287	September
Kent	13.851573	11.578293	October
Kent	9.603741	7.383741	November
Kent	9.031357	7.049387	December
London	6.877389	4.187889	January
London	10.090086	7.559095	February
London	10.263696	7.535699	March
London	11.855089	9.119722	April
London	16.042876	12.880417	May
London	17.622389	14.203722	June
London	19.196882	16.496178	July
London	20.182245	17.435954	August
London	16.380833	13.831164	September
London	13.496962	11.183280	October
London	9.361658	7.030153	November
London	9.010153	6.983217	December
Surrey	6.815958	4.117139	January
Surrey	9.965747	7.492601	February
Surrey	10.155511	7.320175	March
Surrey	11.745722	8.988319	April
Surrey	15.908169	12.741976	May
Surrey	17.478972	13.919569	June
Surrey	19.002198	16.262040	July
Surrey	19.858268	17.079704	August
Surrey	16.133362	13.575474	September
Surrey	13.397177	11.063091	October
Surrey	9.313074	6.941238	November
Surrey	8.912123	6.944945	December
West Sussex	6.901625	4.060194	January
West Sussex	9.983578	7.331997	February
West Sussex	10.040981	7.117164	March
West Sussex	11.657431	8.773028	April
West Sussex	15.522809	12.408562	May
West Sussex	17.061847	13.693375	June
West Sussex	18.629440	16.097888	July
West Sussex	19.532392	16.915282	August
West Sussex	16.108362	13.632543	September
West Sussex	13.788723	11.316358	October
West Sussex	9.848957	7.178985	November
West Sussex	8.913326	6.993807	December
Average	13.078186	10.526951	nan

Reproducibility:

Software:

To download the necessary software:

Use this command in bash or equivalent:

```
$ pip install -r requirements.txt
```

Credentials:

- OpenWeather API
 - You must obtain an OpenWeather API key and store it in the "1_data_collection_code directory" in a text file titled "api_key.txt"
 - We hid our api_key due to security concerns regarding having it on GitHub
- MongoDB
 - You must obtain a MongoDB connection string (not previously used is better so you do not run out of storage) stored in the main directory of the repository in a text file titled "mongo_db_connection_string.txt"
 - A MongoDB account is necessary to get a MongoDB connection string
 - We hid our connection string(s) partially for security, but more so because they are not anonymous

Why PyMongo:

PyMongo was a strong choice for creating our database for several reasons. MongoDB allows the storage of time series data and multiple collections, which improves our storage efficiency. We also preferred to write our code in Python, not only because all of our group members were already familiar with Python, but also because it is frequently used for a variety of different kinds of tasks. This means that we are, theoretically, more likely to reuse the database at some point if it is easily accessible from Python as opposed to, for example, DB Browser. MongoDB is also by far the most used NoSQL management platform^{[\[source\]](#)}, so we figured it may be worth practising more if all else were equal.